

1. Give the Taylor expansion of the function $f(x) = \tan x$ to order 3 then show that

$$\tan x > x + \frac{x^3}{3}, \forall x \in]0, \frac{\pi}{2}[.$$

2. Using Taylor's formula show that

1) $\forall x \geq 0$

$$x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} \leq \ln(1+x) \leq x - \frac{x^2}{2} + \frac{x^3}{3}$$

then give an approximation of $\ln 2$.

2) $\forall x \geq 0$

$$\exp(x) \geq 1 + x + \frac{x^2}{2} + \frac{x^3}{3}.$$

3. Write the Taylor formula to order n at $x_0 = 0$ for the function $f(x) = (1+x)^\alpha$, $\alpha \in \mathbb{R}$.

Using the rules of calculations, deduce the LD of the following functions

$$f_1(x) = \frac{1}{1-x}, f_2(x) = \ln(1-x), f_3(x) = \frac{1}{(1-x)^2}, f_4(x) = \frac{1}{1+x},$$

$$f_5(x) = \ln(1+x), f_6(x) = \arg th(x), f_7(x) = \frac{1}{1+x^2}, f_8(x) = \arctg(x),$$

$$f_9(x) = \frac{1}{\sqrt{1-x^2}}, f_{10}(x) = \arcsin x, f_{11}(x) = \frac{1}{\sqrt{1+x^2}}, f_{12}(x) = \arg sh(x).$$

4. Calculate the LD at the point and order, as indicated.

— $f(x) = \frac{1}{x} \ln(2-x)$ at 1 and to order 2,

— $g(x) = \frac{x-1}{\ln x}$ at 1 and to order 2,

— $h(x) = \ln(\sin x)$ at $\frac{\pi}{2}$ and to order 4,

— $k(x) = \arg sh\left(\frac{1}{x}\right) + \arcsin\left(\frac{1}{x}\right) - \frac{2}{x}$, at $+\infty$ to order 5,

— $l(x) = \frac{\sqrt[3]{1-x}}{x-x^2}$ at 1 to order 3.

5. Determine the asymptotes of curves of the following functions

$$f(x) = x \exp\left(\frac{2x}{1+x}\right), g(x) = \sqrt[3]{x^3 + x + 1}.$$

6. Calculate the limits

$$\lim_{x \rightarrow 0} \frac{2x(1 - \cos x)}{(e^x - 1) \operatorname{tg} x}, \lim_{x \rightarrow 0} \frac{e^x - \exp(\sin x)}{x - \sin x}, \lim_{x \rightarrow +\infty} (1+x)^{\frac{1}{x}},$$
$$\lim_{x \rightarrow +\infty} \left(\frac{2}{\pi} \arctg x - 1\right), \lim_{x \rightarrow 1} (\ln x)^{x-1}, \lim_{x \rightarrow +\infty} x \arctg\left(\frac{1}{1-x}\right).$$

Supplementary exercises

7. Does the function $f(x) = \ln x$ admit a LD near infinity.
8. Give the LD of order 1, 2, 3, 4 near 0 of the function $g(x) = x^4 + x^2 + 2x^3$.
9. Calculate

$$\lim_{x \rightarrow 0} \frac{\exp(\tan(x)) - \exp(x)}{x - \sin(x)}$$

10. Show that for all $x \in [\frac{1}{4}, \frac{1}{2}]$

$$x + \frac{x^3}{8} \leq \arcsin(x) \leq x + \frac{16\sqrt{3}}{9}x^3.$$

11. Let $(a, h) \in \mathbb{R} \times \mathbb{R}_+^*$, $f : [a - h, a + h] \rightarrow \mathbb{R}$ a continuous function over $[a - h, a + h]$, differentiable over $]a - h, a + h[$. Show that there exists $\theta \in]0, 1[$ such that :

$$f(a + h) - f(a - h) = h(f'(a + \theta h) + f'(a - \theta h))$$

12. Let $f(x) = \sqrt[3]{x^3 + 8}$. Determine the equation of the asymptote to the curve of f in the neighborhood of $+\infty$ as well as its position with respect to the curve of the function.

13. Let f be a function defined on $[a, a + h]$, show that the number θ which occurs in the Taylor's Theorem with Lagrange's form of remainder after n terms approaches the limit $1/(n + 1)$ as h approaches zero, provided that $f^{(n+1)}(x)$ is continuous and different from zero at $x = a$.
14. Suppose that f is differentiable on $[0, 1]$ and that $f(0) = f(1) = 0$. Suppose, further, that f'' exists on $(0, 1)$ and is bounded (say $|f''(x)| \leq A$ for $x \in (0, 1)$). Prove that

$$|f'(x)| \leq \frac{A}{2} \quad \text{for } x \in [0, 1].$$

15. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be $2n + 1$ times differentiable on $\mathbb{R}$. Prove that for every $x \in \mathbb{R}$ there is $\theta \in (0, 1)$ such that

$$\begin{aligned} f(x) = & f(0) + \frac{2}{1!} f' \left(\frac{x}{2} \right) \left(\frac{x}{2} \right) + \frac{2}{3!} f^{(3)} \left(\frac{x}{2} \right) \left(\frac{x}{2} \right)^3 \\ & + \cdots + \frac{2}{(2n-1)!} f^{(2n-1)} \left(\frac{x}{2} \right) \left(\frac{x}{2} \right)^{2n-1} \\ & + \frac{2}{(2n+1)!} f^{(2n+1)}(\theta x) \left(\frac{x}{2} \right)^{2n+1}. \end{aligned}$$